Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

September 30, 2024

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Effluent PFAS
Pace Project No.: 50381719

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on September 06, 2024. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Minneapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,

Brian Hall
brian.hall@pacelabs.com
(616)975-4500
Project Manager

Enclosures

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

CERTIFICATIONS

Project: Effluent PFAS
Pace Project No.: 50381719

Pace Analytical Services, LLC - Minneapolis MN

1700 Elm Street SE, Minneapolis, MN 55414
Alabama Certification #: 40770
Alaska Contaminated Sites Certification #: 17-009
Alaska DW Certification #: MN00064
Arizona Certification #: AZ0014
Arkansas DW Certification #: MN00064
Arkansas WW Certification #: 88-0680
California Certification #: 2929
Colorado Certification #: MN00064
Connecticut Certification #: PH-0256
DoD Certification via A2LA #: 2926.01
EPA Region 8 Tribal Water Systems+Wyoming DW Certification #: via MN 027-053-137
Florida Certification #: E87605
Georgia Certification #: 959
GMP+ Certification #: GMP050884
Hawaii Certification #: MN00064
Idaho Certification #: MN00064
Illinois Certification #: 200011
Indiana Certification #: C-MN-01
Iowa Certification #: 368
ISO/IEC 17025 Certification via A2LA #: 2926.01
Kansas Certification #: E-10167
Kentucky DW Certification #: 90062
Kentucky WW Certification #: 90062
Louisiana DEQ Certification #: AI-03086
Louisiana DW Certification #: MN00064
Maine Certification #: MN00064
Maryland Certification #: 322
Michigan Certification #: 9909
Minnesota Certification #: 027-053-137
Minnesota Dept of Ag Approval: via MN 027-053-137
Minnesota Petrofund Registration #: 1240
Mississippi Certification #: MN00064
Missouri Certification #: 10100
Montana Certification #: CERT0092
Nebraska Certification #: NE-OS-18-06
Nevada Certification #: MN00064
New Hampshire Certification #: 2081
New Jersey Certification #: MN002
New York Certification #: 11647
North Carolina DW Certification #: 27700
North Carolina WW Certification #: 530
North Dakota Certification (A2LA) #: R-036
North Dakota Certification (MN) #: R-036
Ohio DW Certification #: 41244
Ohio VAP Certification (1700) #: CL101
Oklahoma Certification #: 9507
Oregon Primary Certification #: MN300001
Oregon Secondary Certification #: MN200001
Pennsylvania Certification #: 68-00563
Puerto Rico Certification #: MN00064
South Carolina Certification #:74003001
Tennessee Certification #: TN02818
Texas Certification #: T104704192
Utah Certification #: MN00064
Vermont Certification #: VT-027053137
Virginia Certification #: 460163
Washington Certification #: C486
West Virginia DEP Certification #: 382
West Virginia DW Certification #: 9952 C
Wisconsin Certification #: 999407970
Wyoming UST Certification via A2LA #: 2926.01
USDA Permit #: P330-19-00208

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SAMPLE SUMMARY

Project: Effluent PFAS
Pace Project No.: 50381719

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50381719001	**Effluent**	Water	09/05/24 11:40	09/06/24 10:30

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SAMPLE ANALYTE COUNT

Project: Effluent PFAS
Pace Project No.: 50381719

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50381719001	**Effluent**	EPA 1633 DRAFT	NBH	64	PASI-M

PASI-M = Pace Analytical Services - Minneapolis

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SUMMARY OF DETECTION

Project: Effluent PFAS
Pace Project No.: 50381719

Lab Sample ID Method	Client Sample ID Parameters	Result	Units	Report Limit	Analyzed	Qualifiers
50381719001	**Effluent**					
EPA 1633 DRAFT	NMeFOSAA	1.2	ng/L	0.99	09/25/24 16:19	
EPA 1633 DRAFT	PFBS	48.8	ng/L	0.99	09/25/24 16:19	
EPA 1633 DRAFT	PFDA	0.39J	ng/L	0.99	09/25/24 16:19	
EPA 1633 DRAFT	PFHxA	12.9	ng/L	0.99	09/25/24 16:19	
EPA 1633 DRAFT	PFBA	4.7	ng/L	4.0	09/25/24 16:19	B
EPA 1633 DRAFT	PFMPA	2.5	ng/L	2.0	09/25/24 16:19	
EPA 1633 DRAFT	PFPeA	3.3	ng/L	2.0	09/25/24 16:19	
EPA 1633 DRAFT	PFHpA	0.50J	ng/L	0.99	09/25/24 16:19	
EPA 1633 DRAFT	PFHxS	0.99J	ng/L	0.99	09/25/24 16:19	
EPA 1633 DRAFT	PFOS	1.6	ng/L	0.99	09/25/24 16:19	
EPA 1633 DRAFT	PFOA	2.3	ng/L	0.99	09/25/24 16:19	

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ANALYTICAL RESULTS

Project: Effluent PFAS
Pace Project No.: 50381719

Sample: Effluent **Lab ID: 50381719001** Collected: 09/05/24 11:40 Received: 09/06/24 10:30 Matrix: Water

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
EPA 1633 DRAFT Water	Analytical Method: EPA 1633 DRAFT Preparation Method: EPA 1633 DRAFT Pace Analytical Services - Minneapolis								
11Cl-PF3OUdS	**<0.77**	ng/L	4.0	0.77	1	09/24/24 14:07	09/25/24 16:19	763051-92-9	
3:3 FTCA	**<2.3**	ng/L	5.0	2.3	1	09/24/24 14:07	09/25/24 16:19	356-02-5	
4:2 FTS	**<0.48**	ng/L	4.0	0.48	1	09/24/24 14:07	09/25/24 16:19	757124-72-4	
5:3 FTCA	**<4.8**	ng/L	24.8	4.8	1	09/24/24 14:07	09/25/24 16:19	914637-49-3	
6:2 FTS	**<0.97**	ng/L	4.0	0.97	1	09/24/24 14:07	09/25/24 16:19	27619-97-2	
7:3 FTCA	**<5.2**	ng/L	24.8	5.2	1	09/24/24 14:07	09/25/24 16:19	812-70-4	
8:2 FTS	**<1.0**	ng/L	4.0	1.0	1	09/24/24 14:07	09/25/24 16:19	39108-34-4	
9Cl-PF3ONS	**<0.82**	ng/L	4.0	0.82	1	09/24/24 14:07	09/25/24 16:19	756426-58-1	
ADONA	**<0.58**	ng/L	4.0	0.58	1	09/24/24 14:07	09/25/24 16:19	919005-14-4	
HFPO-DA	**<0.77**	ng/L	4.0	0.77	1	09/24/24 14:07	09/25/24 16:19	13252-13-6	
NEtFOSAA	**<0.34**	ng/L	0.99	0.34	1	09/24/24 14:07	09/25/24 16:19	2991-50-6	
NEtFOSA	**<0.25**	ng/L	0.99	0.25	1	09/24/24 14:07	09/25/24 16:19	4151-50-2	
NEtFOSE	**<3.2**	ng/L	9.9	3.2	1	09/24/24 14:07	09/25/24 16:19	1691-99-2	
NFDHA	**<0.56**	ng/L	2.0	0.56	1	09/24/24 14:07	09/25/24 16:19	151772-58-6	
NMeFOSAA	**1.2**	ng/L	0.99	0.32	1	09/24/24 14:07	09/25/24 16:19	2355-31-9	
NMeFOSA	**<0.26**	ng/L	0.99	0.26	1	09/24/24 14:07	09/25/24 16:19	31506-32-8	
NMeFOSE	**<2.6**	ng/L	9.9	2.6	1	09/24/24 14:07	09/25/24 16:19	24448-09-7	
PFBS	**48.8**	ng/L	0.99	0.32	1	09/24/24 14:07	09/25/24 16:19	375-73-5	
PFDA	**0.39J**	ng/L	0.99	0.18	1	09/24/24 14:07	09/25/24 16:19	335-76-2	
PFHxA	**12.9**	ng/L	0.99	0.15	1	09/24/24 14:07	09/25/24 16:19	307-24-4	
PFBA	**4.7**	ng/L	4.0	0.61	1	09/24/24 14:07	09/25/24 16:19	375-22-4	B
PFDS	**<0.26**	ng/L	0.99	0.26	1	09/24/24 14:07	09/25/24 16:19	335-77-3	
PFDoS	**<0.27**	ng/L	0.99	0.27	1	09/24/24 14:07	09/25/24 16:19	79780-39-5	
PFEESA	**<0.32**	ng/L	2.0	0.32	1	09/24/24 14:07	09/25/24 16:19	113507-82-7	
PFHpS	**<0.24**	ng/L	0.99	0.24	1	09/24/24 14:07	09/25/24 16:19	375-92-8	
PFMBA	**<0.31**	ng/L	2.0	0.31	1	09/24/24 14:07	09/25/24 16:19	863090-89-5	
PFMPA	**2.5**	ng/L	2.0	0.48	1	09/24/24 14:07	09/25/24 16:19	377-73-1	
PFNS	**<0.24**	ng/L	0.99	0.24	1	09/24/24 14:07	09/25/24 16:19	68259-12-1	
PFOSA	**<0.24**	ng/L	0.99	0.24	1	09/24/24 14:07	09/25/24 16:19	754-91-6	
PFPeA	**3.3**	ng/L	2.0	0.30	1	09/24/24 14:07	09/25/24 16:19	2706-90-3	
PFPeS	**<0.18**	ng/L	0.99	0.18	1	09/24/24 14:07	09/25/24 16:19	2706-91-4	
PFDoA	**<0.22**	ng/L	0.99	0.22	1	09/24/24 14:07	09/25/24 16:19	307-55-1	
PFHpA	**0.50J**	ng/L	0.99	0.21	1	09/24/24 14:07	09/25/24 16:19	375-85-9	
PFHxS	**0.99J**	ng/L	0.99	0.26	1	09/24/24 14:07	09/25/24 16:19	355-46-4	
PFNA	**<0.23**	ng/L	0.99	0.23	1	09/24/24 14:07	09/25/24 16:19	375-95-1	
PFOS	**1.6**	ng/L	0.99	0.17	1	09/24/24 14:07	09/25/24 16:19	1763-23-1	
PFOA	**2.3**	ng/L	0.99	0.34	1	09/24/24 14:07	09/25/24 16:19	335-67-1	
PFTeDA	**<0.28**	ng/L	0.99	0.28	1	09/24/24 14:07	09/25/24 16:19	376-06-7	
PFTrDA	**<0.20**	ng/L	0.99	0.20	1	09/24/24 14:07	09/25/24 16:19	72629-94-8	
PFUnA	**<0.25**	ng/L	0.99	0.25	1	09/24/24 14:07	09/25/24 16:19	2058-94-8	
Surrogates									
13C2-PFDoA (S)	44	%.	10-130		1	09/24/24 14:07	09/25/24 16:19		
13C3HFPO-DA (S)	62	%.	40-130		1	09/24/24 14:07	09/25/24 16:19		
13C3-PFBS (S)	59	%.	40-135		1	09/24/24 14:07	09/25/24 16:19		
13C3-PFHxS (S)	54	%.	40-130		1	09/24/24 14:07	09/25/24 16:19		

REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
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ANALYTICAL RESULTS

Project: Effluent PFAS
Pace Project No.: 50381719

Sample: Effluent **Lab ID: 50381719001** Collected: 09/05/24 11:40 Received: 09/06/24 10:30 Matrix: Water

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
EPA 1633 DRAFT Water	Analytical Method: EPA 1633 DRAFT Preparation Method: EPA 1633 DRAFT								
	Pace Analytical Services - Minneapolis								
Surrogates									
13C4-PFBA (S)	29	%.	5-130		1	09/24/24 14:07	09/25/24 16:19		
13C4-PFHpA (S)	56	%.	40-130		1	09/24/24 14:07	09/25/24 16:19		
13C5-PFHxA (S)	61	%.	40-130		1	09/24/24 14:07	09/25/24 16:19		
13C5-PFPeA (S)	70	%.	40-130		1	09/24/24 14:07	09/25/24 16:19		
13C6-PFDA (S)	50	%.	40-130		1	09/24/24 14:07	09/25/24 16:19		
13C8-PFOA (S)	52	%.	40-130		1	09/24/24 14:07	09/25/24 16:19		
13C8-PFOS (S)	50	%.	40-130		1	09/24/24 14:07	09/25/24 16:19		
13C8-PFOSA (S)	51	%.	40-130		1	09/24/24 14:07	09/25/24 16:19		
13C9-PFNA (S)	51	%.	40-130		1	09/24/24 14:07	09/25/24 16:19		
d3-MeFOSAA (S)	45	%.	40-170		1	09/24/24 14:07	09/25/24 16:19		
d3-NMeFOSA (S)	41	%.	10-130		1	09/24/24 14:07	09/25/24 16:19		
d5-EtFOSAA (S)	45	%.	25-135		1	09/24/24 14:07	09/25/24 16:19		
d5-NEtFOSA (S)	40	%.	10-130		1	09/24/24 14:07	09/25/24 16:19		
d7-NMeFOSE (S)	52	%.	10-130		1	09/24/24 14:07	09/25/24 16:19		
d9-NEtFOSE (S)	49	%.	10-130		1	09/24/24 14:07	09/25/24 16:19		
13C2-PFTA (S)	36	%.	10-130		1	09/24/24 14:07	09/25/24 16:19		
13C7-PFUdA (S)	47	%.	30-130		1	09/24/24 14:07	09/25/24 16:19		
13C24:2FTS (S)	109	%.	40-200		1	09/24/24 14:07	09/25/24 16:19		
13C26:2FTS (S)	85	%.	40-200		1	09/24/24 14:07	09/25/24 16:19		
13C28:2FTS (S)	72	%.	40-300		1	09/24/24 14:07	09/25/24 16:19		

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QUALITY CONTROL DATA

Project: Effluent PFAS
Pace Project No.: 50381719

QC Batch: 968872
QC Batch Method: EPA 1633 DRAFT
Analysis Method: EPA 1633 DRAFT
Analysis Description: 1633 W
Laboratory: Pace Analytical Services - Minneapolis
Associated Lab Samples: 50381719001

METHOD BLANK: 5062889
Matrix: Water
Associated Lab Samples: 50381719001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
11Cl-PF3OUdS	ng/L	<0.73	3.8	0.73	09/25/24 11:22	
3:3 FTCA	ng/L	<2.2	4.7	2.2	09/25/24 11:22	
4:2 FTS	ng/L	<0.45	3.8	0.45	09/25/24 11:22	
5:3 FTCA	ng/L	<4.6	23.6	4.6	09/25/24 11:22	
6:2 FTS	ng/L	<0.92	3.8	0.92	09/25/24 11:22	
7:3 FTCA	ng/L	<4.9	23.6	4.9	09/25/24 11:22	
8:2 FTS	ng/L	<0.99	3.8	0.99	09/25/24 11:22	
9Cl-PF3ONS	ng/L	<0.78	3.8	0.78	09/25/24 11:22	
ADONA	ng/L	<0.55	3.8	0.55	09/25/24 11:22	
HFPO-DA	ng/L	<0.73	3.8	0.73	09/25/24 11:22	
NEtFOSA	ng/L	<0.24	0.94	0.24	09/25/24 11:22	
NEtFOSAA	ng/L	<0.33	0.94	0.33	09/25/24 11:22	
NEtFOSE	ng/L	<3.0	9.4	3.0	09/25/24 11:22	
NFDHA	ng/L	<0.53	1.9	0.53	09/25/24 11:22	
NMeFOSA	ng/L	<0.25	0.94	0.25	09/25/24 11:22	
NMeFOSAA	ng/L	<0.30	0.94	0.30	09/25/24 11:22	
NMeFOSE	ng/L	<2.5	9.4	2.5	09/25/24 11:22	
PFBA	ng/L	0.97J	3.8	0.58	09/25/24 11:22	
PFBS	ng/L	<0.30	0.94	0.30	09/25/24 11:22	
PFDA	ng/L	<0.17	0.94	0.17	09/25/24 11:22	
PFDoA	ng/L	<0.21	0.94	0.21	09/25/24 11:22	
PFDoS	ng/L	<0.26	0.94	0.26	09/25/24 11:22	
PFDS	ng/L	<0.25	0.94	0.25	09/25/24 11:22	
PFEESA	ng/L	<0.30	1.9	0.30	09/25/24 11:22	
PFHpA	ng/L	<0.20	0.94	0.20	09/25/24 11:22	
PFHpS	ng/L	<0.23	0.94	0.23	09/25/24 11:22	
PFHxA	ng/L	<0.14	0.94	0.14	09/25/24 11:22	
PFHxS	ng/L	<0.24	0.94	0.24	09/25/24 11:22	
PFMBA	ng/L	<0.30	1.9	0.30	09/25/24 11:22	
PFMPA	ng/L	<0.46	1.9	0.46	09/25/24 11:22	
PFNA	ng/L	<0.22	0.94	0.22	09/25/24 11:22	
PFNS	ng/L	<0.23	0.94	0.23	09/25/24 11:22	
PFOA	ng/L	<0.32	0.94	0.32	09/25/24 11:22	
PFOS	ng/L	<0.16	0.94	0.16	09/25/24 11:22	
PFOSA	ng/L	<0.22	0.94	0.22	09/25/24 11:22	
PFPeA	ng/L	<0.28	1.9	0.28	09/25/24 11:22	
PFPeS	ng/L	<0.17	0.94	0.17	09/25/24 11:22	
PFTeDA	ng/L	<0.27	0.94	0.27	09/25/24 11:22	
PFTrDA	ng/L	<0.19	0.94	0.19	09/25/24 11:22	
PFUnA	ng/L	<0.24	0.94	0.24	09/25/24 11:22	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Effluent PFAS
Pace Project No.: 50381719

METHOD BLANK: 5062889 Matrix: Water
Associated Lab Samples: 50381719001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
13C2-PFDoA (S)	%.	88	10-130		09/25/24 11:22	
13C2-PFTA (S)	%.	88	10-130		09/25/24 11:22	
13C24:2FTS (S)	%.	91	40-200		09/25/24 11:22	
13C26:2FTS (S)	%.	94	40-200		09/25/24 11:22	
13C28:2FTS (S)	%.	87	40-300		09/25/24 11:22	
13C3-PFBS (S)	%.	99	40-135		09/25/24 11:22	
13C3-PFHxS (S)	%.	95	40-130		09/25/24 11:22	
13C3HFPO-DA (S)	%.	94	40-130		09/25/24 11:22	
13C4-PFBA (S)	%.	95	5-130		09/25/24 11:22	
13C4-PFHpA (S)	%.	93	40-130		09/25/24 11:22	
13C5-PFHxA (S)	%.	93	40-130		09/25/24 11:22	
13C5-PFPeA (S)	%.	97	40-130		09/25/24 11:22	
13C6-PFDA (S)	%.	91	40-130		09/25/24 11:22	
13C7-PFUdA (S)	%.	92	30-130		09/25/24 11:22	
13C8-PFOA (S)	%.	91	40-130		09/25/24 11:22	
13C8-PFOS (S)	%.	95	40-130		09/25/24 11:22	
13C8-PFOSA (S)	%.	84	40-130		09/25/24 11:22	
13C9-PFNA (S)	%.	93	40-130		09/25/24 11:22	
d3-MeFOSAA (S)	%.	79	40-170		09/25/24 11:22	
d3-NMeFOSA (S)	%.	74	10-130		09/25/24 11:22	
d5-EtFOSAA (S)	%.	77	25-135		09/25/24 11:22	
d5-NEtFOSA (S)	%.	78	10-130		09/25/24 11:22	
d7-NMeFOSE (S)	%.	85	10-130		09/25/24 11:22	
d9-NEtFOSE (S)	%.	87	10-130		09/25/24 11:22	

LABORATORY CONTROL SAMPLE & LCSD: 5062890 5062891

Parameter	Units	Spike Conc.	LCS Result	LCSD Result	LCS % Rec	LCSD % Rec	% Rec Limits	RPD	Max RPD	Qualifiers
11Cl-PF3OUdS	ng/L	88.3	90.3	86.9	102	99	55-160	4	30	
3:3 FTCA	ng/L	117	125	120	107	103	65-130	4	30	
4:2 FTS	ng/L	87.7	91.4	90.7	104	104	70-145	1	30	
5:3 FTCA	ng/L	584	564	531	96	91	70-135	6	30	
6:2 FTS	ng/L	88.8	95.5	94.0	108	106	65-155	2	30	
7:3 FTCA	ng/L	584	551	522	94	90	50-145	5	30	
8:2 FTS	ng/L	90	100	98.3	111	110	60-150	2	30	
9Cl-PF3ONS	ng/L	87.7	91.7	86.9	105	99	70-155	5	30	
ADONA	ng/L	88.3	90.2	85.7	102	98	65-145	5	30	
HFPO-DA	ng/L	93.5	98.9	92.3	106	99	70-140	7	30	
NEtFOSA	ng/L	23.4	22.7	21.7	97	93	65-145	4	30	
NEtFOSAA	ng/L	23.4	23.0	22.1	98	95	70-145	4	30	
NEtFOSE	ng/L	234	245	237	105	102	70-135	3	30	
NFDHA	ng/L	46.8	48.0	48.7	103	105	50-150	2	30	
NMeFOSA	ng/L	23.4	22.9	21.7	98	93	60-150	5	30	
NMeFOSAA	ng/L	23.4	21.8	21.8	93	93	50-140	0	30	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: Effluent PFAS
Pace Project No.: 50381719

LABORATORY CONTROL SAMPLE & LCSD: 5062890 5062891

Parameter	Units	Spike Conc.	LCS Result	LCSD Result	LCS % Rec	LCSD % Rec	% Rec Limits	RPD	Max RPD	Qualifiers
NMeFOSE	ng/L	234	244	241	104	103	70-145	1	30	
PFBA	ng/L	93.5	97.0	93.8	104	101	70-140	3	30	
PFBS	ng/L	20.7	21.2	20.2	102	98	60-145	5	30	
PFDA	ng/L	23.4	23.2	24.1	99	103	70-140	4	30	
PFDoA	ng/L	23.4	24.0	23.0	102	99	70-140	4	30	
PFDoS	ng/L	22.7	20.6	20.2	91	89	50-145	2	30	
PFDS	ng/L	22.6	22.3	21.3	99	95	60-145	4	30	
PFEESA	ng/L	41.6	42.9	43.0	103	104	70-140	0	30	
PFHpA	ng/L	23.4	24.4	23.7	104	102	70-150	3	30	
PFHpS	ng/L	22.3	23.2	21.9	104	99	70-150	6	30	
PFHxA	ng/L	23.4	24.2	23.3	103	100	70-145	4	30	
PFHxS	ng/L	21.4	21.5	20.8	101	98	65-145	3	30	
PFMBA	ng/L	46.8	46.3	45.8	99	98	60-150	1	30	
PFMPA	ng/L	46.8	47.1	47.1	101	101	55-140	0	30	
PFNA	ng/L	23.4	23.9	23.2	102	100	70-150	3	30	
PFNS	ng/L	22.5	21.9	21.0	97	94	65-145	4	30	
PFOA	ng/L	23.4	23.5	22.8	101	98	70-150	3	30	
PFOS	ng/L	21.7	21.7	20.1	100	93	55-150	8	30	
PFOSA	ng/L	23.4	23.3	22.9	100	98	70-145	2	30	
PFPeA	ng/L	46.8	48.0	47.0	103	101	65-135	2	30	
PFPeS	ng/L	22	22.5	22.1	103	101	65-140	2	30	
PFTeDA	ng/L	23.4	24.8	23.3	106	100	60-140	6	30	
PFTrDA	ng/L	23.4	23.6	23.1	101	99	65-140	2	30	
PFUnA	ng/L	23.4	23.0	23.1	98	99	70-145	0	30	
13C2-PFDoA (S)	%.				92	91	10-130			
13C2-PFTA (S)	%.				90	91	10-130			
13C24:2FTS (S)	%.				91	92	40-200			
13C26:2FTS (S)	%.				95	98	40-200			
13C28:2FTS (S)	%.				91	92	40-300			
13C3-PFBS (S)	%.				104	103	40-135			
13C3-PFHxS (S)	%.				102	102	40-130			
13C3HFPO-DA (S)	%.				99	101	40-130			
13C4-PFBA (S)	%.				97	98	5-130			
13C4-PFHpA (S)	%.				97	98	40-130			
13C5-PFHxA (S)	%.				97	96	40-130			
13C5-PFPeA (S)	%.				101	100	40-130			
13C6-PFDA (S)	%.				95	92	40-130			
13C7-PFUdA (S)	%.				97	93	30-130			
13C8-PFOA (S)	%.				99	96	40-130			
13C8-PFOS (S)	%.				99	102	40-130			
13C8-PFOSA (S)	%.				89	88	40-130			
13C9-PFNA (S)	%.				97	97	40-130			
d3-MeFOSAA (S)	%.				86	85	40-170			
d3-NMeFOSA (S)	%.				80	82	10-130			
d5-EtFOSAA (S)	%.				84	87	25-135			
d5-NEtFOSA (S)	%.				82	87	10-130			
d7-NMeFOSE (S)	%.				87	89	10-130			

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Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: Effluent PFAS
Pace Project No.: 50381719

LABORATORY CONTROL SAMPLE & LCSD: 5062890 5062891

Parameter	Units	Spike Conc.	LCS Result	LCSD Result	LCS % Rec	LCSD % Rec	% Rec Limits	RPD	Max RPD	Qualifiers
d9-NEtFOSE (S)	%.				89	92	10-130			

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALIFIERS

Project:	Effluent PFAS
Pace Project No.:	50381719

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.

ND - Not Detected at or above adjusted reporting limit.

TNTC - Too Numerous To Count

J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.

MDL - Adjusted Method Detection Limit.

PQL - Practical Quantitation Limit.

RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.

S - Surrogate

1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.

Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.

LCS(D) - Laboratory Control Sample (Duplicate)

MS(D) - Matrix Spike (Duplicate)

DUP - Sample Duplicate

RPD - Relative Percent Difference

NC - Not Calculable.

SG - Silica Gel - Clean-Up

U - Indicates the compound was analyzed for, but not detected.

N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.

Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.

Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.

TNI - The NELAC Institute.

ANALYTE QUALIFIERS

B Analyte was detected in the associated method blank.

Pace Analytical Services, LLC
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Grand Rapids, MI 49512
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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Effluent PFAS
Pace Project No.: 50381719

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50381719001	**Effluent**	EPA 1633 DRAFT	968872	EPA 1633 DRAFT	970340

Pace® Location Requested (City/State):
Pace Analytical Grand Rapids
4171 40th Street SE, Grand Rapids, MI 49512

CHAIN-OF-CUSTODY Analytical Request Document

Chain-of-Custody is a LEGAL DOCUMENT - Complete all relevant fields

LAB USE ONLY- Affix Workorder/Login Label Here

WO#: 50381719

50381719

Company Name:	City of Cadillac
Street Address:	1121 Plett Rd Cadillac, MI 49601
Customer Project #:	
Project Name:	Effluent PFAS
Site Collection Info/Facility ID (as applicable):	
Contact/Report To:	Cindy Tomaszewski
Phone #:	231-775-2368
E-Mail:	ctomaszewski@cadillac-mi.net
Cc E-Mail:	
Invoice To:	Cindy Tomaszewski
Invoice E-Mail:	ctomaszewski@cadillac-mi.net
Purchase Order # (if applicable):	
Quote #:	
County / State origin of sample(s):	Michigan

Time Zone Collected: [ ] AK [ ] PT [ ] MT [ ] CT [ ] ET

Data Deliverables: [ ] Level II [ ] Level III [ ] Level IV [ ] EQUIS [ ] Other

Regulatory Program (DW, RCRA, etc.) as applicable: EGLE-NPDES Reportable [X] Yes [ ] No

Rush (Pre-approval required): [ ] Same Day [ ] 1 Day [ ] 2 Day [ ] 3 Day [ ] Other ______

DW PWSID # or WW Permit # as applicable: MI0020257

Date Results Requested:

Field Filtered (if applicable): [ ] Yes [X] No

Analysis:

* Matrix Codes (Insert in Matrix box below): Drinking Water (DW), Ground Water (GW), Waste Water (WW), Product (P), Soil/Solid (SS), Oil (OL), Wipe (WP), Tissue (TS), Bioassay (B), Vapor (V), Surface Water (SW), Sediment (SED), Sludge (SL), Caulk (CK), Leachate (LL), Biosolid (BS), Other (OT)

Specify Container Size **: 2&4

Identify Container Preservative Type***: 1

Analysis Requested

**Container Size: (1) 1L, (2) 500mL, (3) 250mL, (4) 125mL, (5) 100mL, (6) 40mL vial, (7) EnCore, (8) TerraCore, (9) 90mL, (10) Other

*** Preservative Types: (1) None, (2) HNO3, (3) H2SO4, (4) HCl, (5) NaOH, (6) Zn Acetate, (7) NaHSO4, (8) Sod. Thiosulfate, (9) Ascorbic Acid, (10) MeOH, (11) Other

Lab Use Only

Proj. Mgr:	**Brian Hall**
AcctNum / Client ID:	
Table #:	
Profile / Template:	**7675**
Prelog / Bottle Ord. ID:	**EZ 3124085**

Preservation non-conformance identified for sample.

Customer Sample ID	Matrix *	Comp / Grab	Composite Start Date	Composite Start Time	Collected or Composite End Date	Collected or Composite End Time	# Cont.	Res. Chlorine Results	Res. Chlorine Units	1633 PFAS - Pace MN	Sample Comment
Effluent	WW	G	—	—	9-5-2024	1140	3	N/A	—	X	

Additional Instructions from Pace®:
1633 PFAS - 40 Compounds
****Requires Reporting by 10th of Following Month****

Collected By: (Printed Name) CINDY TOMASZEWSKI
Signature: CTomaszewski

Customer Remarks / Special Conditions / Possible Hazards: Need RL = 2.0 ng/L

# Coolers:	Thermometer ID:	Correction Factor (°C):	Obs. Temp. (°C)	Corrected Temp. (°C)	On Ice:

Relinquished by/Company: (Signature)	Date/Time:	Received by/Company: (Signature)	Date/Time:
CTomaszewski / City of Cadillac	9-5-24 / 1330	UPS Ground Shipping	
		[illegible] / Pace	9/6/24 1030

Tracking Number:

Delivered by: [ ] In- Person [ ] Courier [ ] FedEX [ ] UPS [ ] Other

Page: 1 of 1

Submitting a sample via this chain of custody constitutes acknowledgment and acceptance of the Pace® Terms and Conditions found at https://www.pacelabs.com/resource-library/resource/pace-terms-and-conditions/

ENV-FRM-CORQ-0019_v02_110123 ©

Pace

Sample Conditions Upon Receipt Form (SCUR)

Date/Time: 9/6 11 10	**Evaluated By:** DAS	**WO#: 50381719** PM: BJH Due Date: 10/07/24 CLIENT: GR-Cadillac	
Client: City of Cadillac	**PM:** BJH		
Lab Notified of Rush or Short Holds: YES NO NA			

Project Received Via: FedEx UPS Client Pace Courier Other: ____________

				Comments:
Custody Seal Present and Intact:	YES	NO	NA	
Received Sample Information Form (SIF): Drinking Waters Only	YES	NO	NA	
Short Hold Present (≤ 48 Hours):	YES	NO		
Sample Received in Hold:	YES	NO		
Custody Signature Present:	YES	NO		
Collector Signature Present:	YES	NO		
Sample Collected Today and On Ice:	YES	NO	N/A	
IR Gun #: 350 351 **Therm #:** 282 283	**Temp. should be 0°C - 6°C (Initial/Corrected)**			
Ice Type: WET Bagged / WET Loose BLUE NONE	**1.** Cooler Temp. Upon Receipt: 1.7/1.5 °C			
Ice Location: TOP BOTTOM MIDDLE DISPERSED	**2.** Cooler Temp. Upon Receipt: ____________ °C			
Temp Blank Received:	YES	NO		
Sample Label Matches COC (ID/Date/Time):	YES	NO		
Container Intact:	YES	NO		
Correct Container:	YES	NO		
Sufficient Volume:	YES	NO		
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation. Denote with red dot on container lid if unacceptable. **pH Strip Lot #:** ____________ *Exceptions are VOA, coliform, LLHg, O&G/TPH, or any container with a septum cap or preserved with HCl*	YES	NO	N/A	
Residual Chlorine Absent: **Cl_2 Strip Lot #:** ____________ *Applies to SVOC 625, PCB/Pest. 608, Total/Amenable/Available/Free Cyanide*	YES	NO	N/A	
VOA Headspace Acceptable (<6mm): Denote with silver x on rim of container cap if unacceptable.	YES	NO	N/A	
Trip Blank Received: HCl MeOH Other: ____________	YES	NO	ON HOLD	

Comments:	**3.** Cooler Temp. Upon Receipt: ____________ °C
	4. Cooler Temp. Upon Receipt: ____________ °C
	Non-Conformance Form Required: YES NO